Name: ____________________

In-class Quiz
#04: Shift, Rotate and Logical Operations

Shift and Rotate Operations Codes

What is the hexadecimal result of the following shift operation (hexadecimal value in R0)? And what are the values of condition flags? Assume all the flags are cleared before the operations. (Show your work.) (6 PTS each)

1.	`MOV32 R1, #0x8282D0D0`
	`ASRS R0, R1, #2`
	ANS: ____________________.
	N: _______; Z: ________; C: ________.

2.	`MOV32 R1, #0xABEF1234`
	`MOV R2, #0x3`
	`LSRS R0, R1, R2`
	ANS: ____________________.
	N: _______; Z: ________; C: ________.

3.	`MOV32 R1, #0x9876ABCD`
	`MOV R2, #0x4`
	`LSLS R0, R1, R2`
	ANS: ____________________.
	N: _______; Z: ________; C: ________.

4.	`MOV32 R1, #0xDD22EE77`
	`MOV R2, #0x4`
	`RORS R0, R1, R2`
	ANS: ______________________________.
	N: _______; Z: ________; C: ________.

Logical Operations

(6 PTS each)

5. What is the hexadecimal result of the following **OR** operation? (Show your work.)

   #0xA3 | #0x56 = __________________

6. What is the hexadecimal result of the following **AND** operation? (Show your work.)

   #0x9F & #0xA7 = __________________

7. What is the hexadecimal result of the following **EOR** (exclusive OR) operation? (Show your work.)

   #0x4C ^ #0x78 = __________________

Operand2

You already know the flexible second operand (oprand2) in ARM assembly language support optionally shift. For example, the instruction

```
MOV  R2, R1, LSL #2
```

will firstly logical shift left the value in register R1, and MOV the shifted value into R2.

8. (7 PTS) Please predict the hexadecimal result of the following code (hexadecimal value in R0)? And what are the values of condition flags? Assume all the flags are cleared before the operations. (Show your work.)

```
MOV32   R1, #0x0022EE77
MOV32   R2, #0x30000000
ADDS    R0, R1, R2, LSL #2
```

ANS: ______________________________.

N: _______; Z: ________; C: ________; V: ________;

9. (6 PTS) Use the flexible second operand to write a single assembly instruction to calculate

$$R0 = R1 - R2/64$$

ANS: ____.

Logical Masks

(15 PTS each)

10. Assume you want to use a logical operation to guarantee that **bits 13, 12, 7, and 6** in the **R0** register are cleared and then store the result back in **R0** (32 bits, 0 to 31). 1) If you use the AND operation, what mask value would you load into **R3** so that the assembly instruction given would accomplish this (i.e., **bits 13, 12, 7, & 6 clear**)? Give your answer in **hexadecimal**. 2) If you use the BIC operation, what the mask will be.

1) `AND R0, R0, R3`

ANS: R3 = _______________________________ (mask in hexadecimal)

2) `BIC R0, R0, R3`

ANS: R3 = ______________________________ (mask in hexadecimal)

11. Assume you want to use a logical operation to guarantee that **bits 19, 18, 17, and 16** in this register are set high and then store the result back in **R0**. Remembering that the ORN operation can be used to set individual bits, what mask value would you load into **R3** so that the assembly instruction given would accomplish this (i.e., **bits 19, 18, 17, & 16** set)? If the value in **R0** is shown as in the following before the instruction, what is the value in **R0** after the instruction.

```
ORN R0, R0, R3
```

ANS: R3 = ______________________________ (mask in hexadecimal)

If R0 = **1010.1011.0010.0011.1100.1101.0100.0101** before the instruction
What is
R0 = ________________________________(in hexadecimal after the instruction).

12. Write the values in register R0 or R3 after the instruction

```
EOR R0, R0, R3
```

If

1) R0 = 0x3456ABCD and R3 = 0xFF000000before the instruction

What is

R0 = __________________________________ (in hexadecimal after the instruction),

2) R0 = 0x3456ABCD before the instruction, and R0 = 0x3A5654CD after the instruction, what is the mask

R3 = __________________________________ (in hexadecimal).